

Descriptive Evidence Appendix to “Automation, Capital Taxation, and Public Insurance”

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Contents

D1	Historical episodes and fiscal state variables	2
D1.1	Purpose and evidence status	2
D1.2	Four distinct empirical objects	2
D1.3	United States, 1990–2007	2
D1.4	Germany, 1994–2014	3
D1.5	Germany, 2004–07	3
D1.6	Asset payoffs in worker-squeeze states	4
D1.7	Long-run taxes, equity, and wealth	4
D1.8	A recent robotics claim and its funding	7
D1.9	Mapping evidence to the models	7
D1.10	Remaining empirical work	7

D1 Historical episodes and fiscal state variables

D1.1 Purpose and evidence status

This appendix documents that labour income, asset values, tax receipts, transfers, and government financing do not move as a single sufficient statistic. That fact motivates a model in which workers, the fiscal authority, and market investors value resources differently across states. The evidence does not identify automation shocks, estimate either stochastic discount factor, or calibrate the probability of a labour-free successor economy.

The United States from 1990 to 2007 and Germany from 1994 to 2014 follow windows used in the robot-exposure literature. Germany from 2004 to 2007 is a close-up chosen inside the longer window because it displays a clear divergence between worker and capital income. It is not an independently identified technology event. The figures reproduce existing project outputs whose complete release lineage is not yet frozen. They are appropriate for this internal draft and for designing the empirical work, not for a causal or publication-stage claim.

Source record. The immediate synthesis is the project’s note *Descriptive automation, worker, fiscal, and portfolio intuitions*; the plots come from the dated long-history output bundle of 3 September 2026. The underlying records are the long-history panel, OECD episode-card, three-country asset-accounting, and revenue-and-transfer construction memos. The v6 manuscript manifest records their exact file locators. These records were checked for this internal draft but have not yet been frozen as a public replication release. All historical numbers and figures in this volume should therefore be read as provisional descriptive evidence.

D1.2 Four distinct empirical objects

An automation *level* is the installed stock or penetration of a technology. An automation *acceleration* is a period in which exposure grows unusually quickly. Automation-like *redistribution* is a shift from labour toward capital income that may also reflect globalization, bargaining, composition, policy, or the cycle. A traded robotics claim is an asset payoff of technology-producing firms. It is neither a technology quantity nor an instrument for identifying an automation shock.

D1.3 United States, 1990–2007

During the US robot-exposure window, the national labour share fell by about 0.07 percentage points per year. Aggregate real compensation nevertheless grew by about 2.35 percent per year, real compensation per employee by about 0.9 percent, and a separately constructed real hourly-compensation series by about 1.3 percent. US equity returned roughly 9 percent per year in real terms, a long rest-of-world claim about 7.3 percent, and a broad US business-equipment producer portfolio about 13.4 percent.

Table D1: Distinct objects in the historical exercise

Object	Current proxy	Use	Limit
Automation level	Candidate ICT, software, and adoption series	Technology penetration	No common long-run measure is frozen.
Automation exposure	US 1990–2007 and Germany 1994–2014	Literature-timed windows	Not a national causal effect.
Redistribution state	Labour-share and worker-income movements	Shared growth versus worker squeeze	Automation is not the unique cause.
Traded payoff	Domestic, external, producer, and robotics equity	Resources available through assets	Not an automation measure or premium.

This is the shared-growth case. Capital income and asset values performed strongly and labour’s share declined, but average worker compensation still rose. Local exposure estimates can coexist with this aggregate pattern because national data include less-exposed places, sectoral reallocation, entry, and aggregate demand. The episode illustrates unequal upside: a public capital claim can share gains with workers even when workers are not absolutely worse off. It does not show that robot exposure caused the national compensation or asset paths.

D1.4 Germany, 1994–2014

Germany’s longer robot-adjustment window lies between shared growth and a worker squeeze. The national labour share fell by about 0.16 percentage points per year. Aggregate real worker compensation grew by only about 0.66 percent; compensation per employee was approximately flat or slightly negative, while hourly compensation was slightly positive. Domestic real equity returned roughly 8–11 percent per year across two available constructions.

The national pattern is consistent with manufacturing displacement being offset by service employment and other margins. It does not translate a local employment effect into national welfare or show that robots caused the equity return. Its value is the aggregation lesson: one episode can contain large losses for exposed workers, small average per-worker gains, and high capital returns.

D1.5 Germany, 2004–07

The 2004–07 close-up is the clearest current example of a worker squeeze combined with a capital boom. Labour’s share fell by about 0.97 percentage points per year. Aggregate real worker compensation fell by about 0.54 percent, compensation per employee by about 1.20 percent, and hourly compensation by about 1.51 percent. A capital-tax-base proxy grew by about 3.72 percent. Domestic real equity returned roughly 12.5–18 percent per year across two constructions, and the long external claim about 9.3 percent.

The fiscal accounts do not describe the same state as the worker accounts. Relative to 2000–

03, harmonized total taxes fell only 0.34 percentage points of GDP in 2004–07. Personal income tax fell 0.78 points, employee contributions 0.37 points, and employer contributions 0.54 points; corporate income tax rose 0.75 points. On a broader national-accounts measure, receipts fell about 1.33 points. Transfers fell about 0.65 points, interest costs 0.30 points, the deficit improved by 1.08 points, and debt-ratio growth slowed.

The appropriate description is worker squeeze plus capital boom without a simultaneous Treasury crisis. Worker marginal value may have been high even though consolidated public financing was not acutely stressed. Transfers cancel in consolidated resources but remain central to household liquidity and Treasury cash. Corporate receipts can rise while labour-based receipts fall, and an equity revaluation is not current cash.

D1.6 Asset payoffs in worker-squeeze states

In the longer return construction, US domestic equity averages about 15.3 percent real in nine capital/profit-boom observations and 1.9 percent in six worker-squeeze observations; external-return means are about 15.7 and -1.7 percent. In the United Kingdom, domestic-equity means are about 19.0 and 6.2 percent, while external means are 13.8 and -1.2 percent, across nine and five observations.

These cells are small. Domestic equity is negative in roughly half the US squeeze years and 40 percent of the UK squeeze years. The descriptive fact is deterioration, not that home equity reliably hedges workers. A debt-financed public position may be fragile if the asset falls when tax receipts weaken or transfers rise. The empirical target is a joint state: worker resources, fiscal cash flow, asset payoff, and funding cost.

D1.7 Long-run taxes, equity, and wealth

Over the endpoints of the harmonized OECD panel, total tax revenue rises from about 23.6 to 25.6 percent of GDP in the United States, 30.1 to 34.4 percent in the United Kingdom, and 31.7 to 38.0 percent in Germany. Over these endpoints, aggregate tax share does not fall in any of the three countries. This does not rule out fiscal vulnerability: tax composition, expenditure demands, and financing conditions can move differently.

Long-run equity and wealth series show the scale of resources a sufficiently early public holding could have captured. They do not show that the position was financeable, well governed, or aligned with worker stress. Total returns also combine distributions with revaluation, which have different implications for current transfers and net worth.

Robot-era episodes can combine similar labour-share declines with very different worker outcomes
National descriptive averages; the 2004–07 panel is a close-up within the German episode

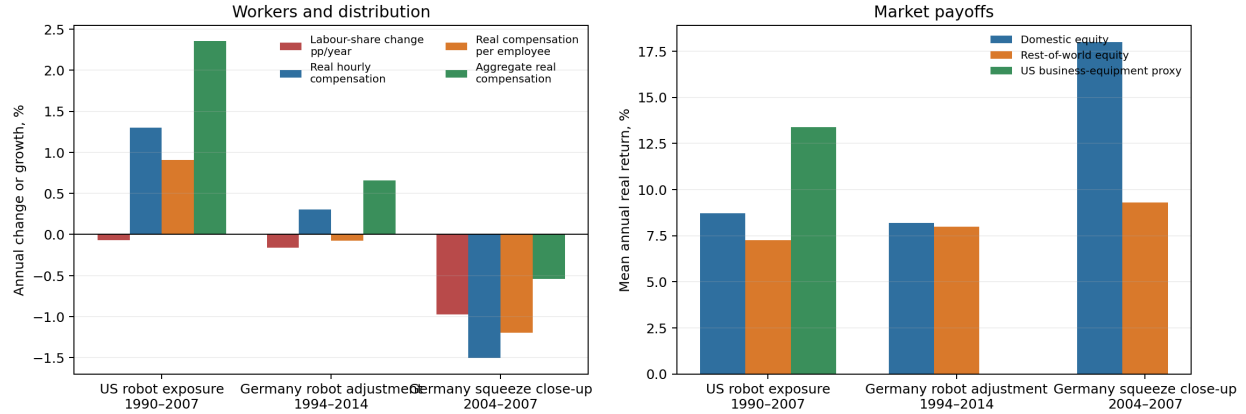


Figure D1: Worker and asset outcomes in selected episodes. This existing output is based on provisional descriptive constructions. The US 1990–2007 and Germany 1994–2014 windows are literature motivated; Germany 2004–07 is a selected subperiod. The figure does not identify national automation effects or estimate insurance value.

Total receipts can hide a shift between labour-linked and corporate tax bases
Descriptive pre-period versus during-period differences

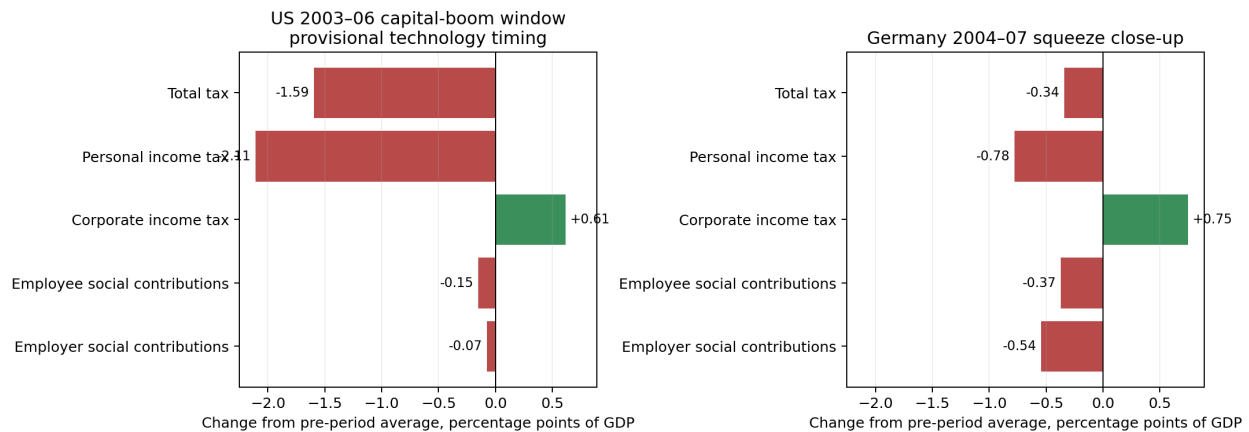


Figure D2: Tax-composition changes in selected periods. Series and timing remain provisional. The exhibit separates personal, payroll, corporate, and other receipts; it is not a causal response to automation.

Equity returns depend on the worker state, not just on a falling labour share
1951–2024; state definitions fixed by the earlier labour-share analysis

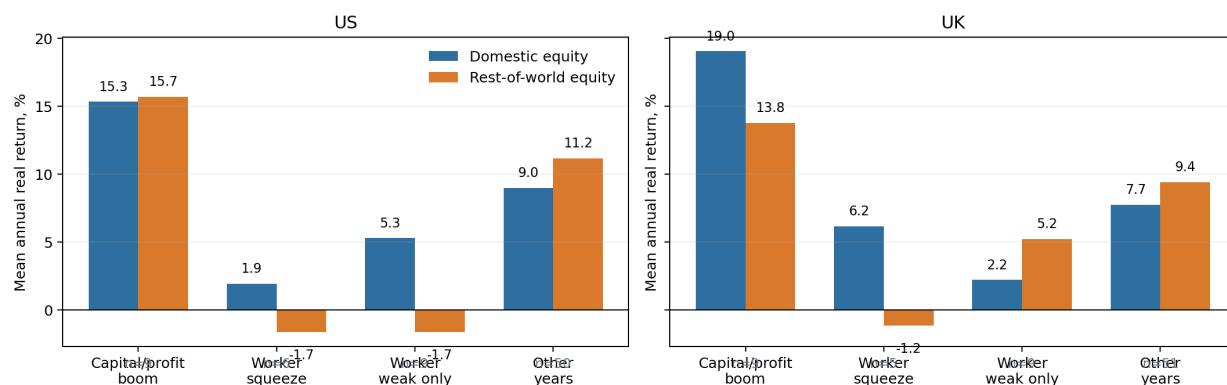


Figure D3: Real asset returns across provisional worker and fiscal states. Sample sizes are small and the state definitions are descriptive. The figure motivates joint-state measurement; it does not establish a reliable hedge.

Both domestic and foreign equity delivered large long-run gains—but this says nothing about when they paid
1950–2020, buy-and-hold real wealth

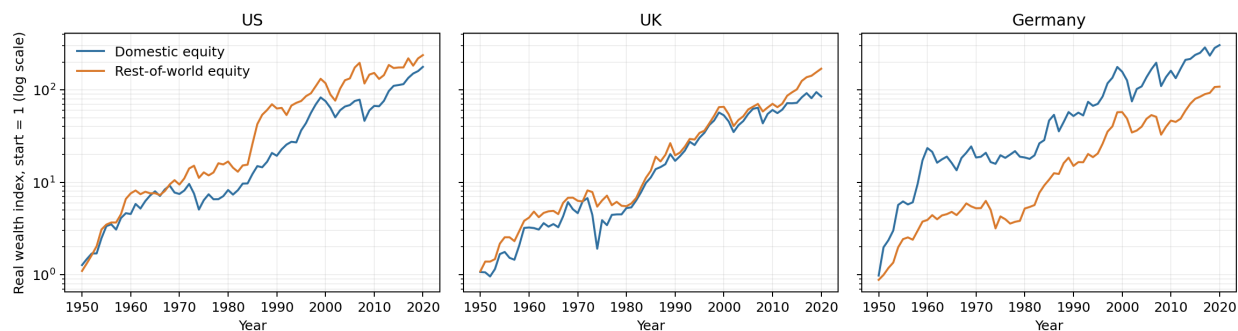


Figure D4: Long-run equity and wealth indicators from the provisional output package. These series provide scale and timing context, not a feasible sovereign-wealth counterfactual.

D1.8 A recent robotics claim and its funding

The recent robotics-equity sample contains only twelve full annual observations. The claim returned about 43.5 percent in real US terms in 2020 and -38.8 percent in 2022. The same security returned about 39.9 percent in sterling and 33.1 percent in euros in 2020 because exchange rates and home inflation alter each government's payoff.

Simple robotics-on-domestic-market betas are about 1.28 for the United States, 0.60 for the United Kingdom, and 0.41 for Germany. Residual volatility is large. These estimates do not establish an automation premium, and the three currency returns are not three independent assets. They show why a technology-equity proposal must specify currency, funding, distributions, and rebalancing.

A pre-existing holding can deliver exceptional upside when workers are stressed, as in 2020, and a severe drawdown when real funding costs rise, as in 2022. A levered position then combines asset loss with expensive refinancing. Expected return does not remove this bad-leverage state.

D1.9 Mapping evidence to the models

The continuous-risk model needs at least two innovations because one aggregate shock cannot reproduce the historical distinctions. One direction can raise output and asset values while lowering labour's share. Another can weaken workers and the fiscal base or alter relative domestic and external payoffs. The representation is not estimated from these episodes; it is the minimum state space needed to ask which component a claim reaches.

The event-arrival model addresses a different uncertainty. Historical episodes show partial adjustment in economies where labour remains productive. They do not reveal a labour-free scalable-capital economy. The AK successor is a conditional stress state, not an extrapolation from Germany or the United States. The fixed-policy reduced-form exercise lies between them: it imposes a factor-income path and evaluates announced fiscal rules, but does not reproduce the Ramsey capital mechanism structurally.

D1.10 Remaining empirical work

A paper-ready design should freeze episode definitions; report means, medians, negative-return frequencies, and uncertainty for each claim; and estimate descriptive panel relationships using labour-share decline, worker weakness, and their interaction. It should repeat the design separately for total receipts, personal and corporate tax, social contributions, transfers, interest, the balance, and debt.

Asset accounting should separate dividends, revaluation, exchange rates, inflation, and funding carry. Event plots for US 1990–2007 and Germany 1994–2014 should be prespecified; Germany 2004–07 should remain an illustrative subperiod. Interpretation should be reopened if a clean

adoption measure does not reproduce the state split, if worker-squeeze payoffs disappear under alternative return constructions, or if apparent fiscal insulation is an accounting-perimeter artifact.

Until then, the evidence supports the paper's state distinctions and its demand for a joint worker-fiscal-asset account. It does not supply causal identification, event probabilities, portfolio weights, or policy magnitudes.

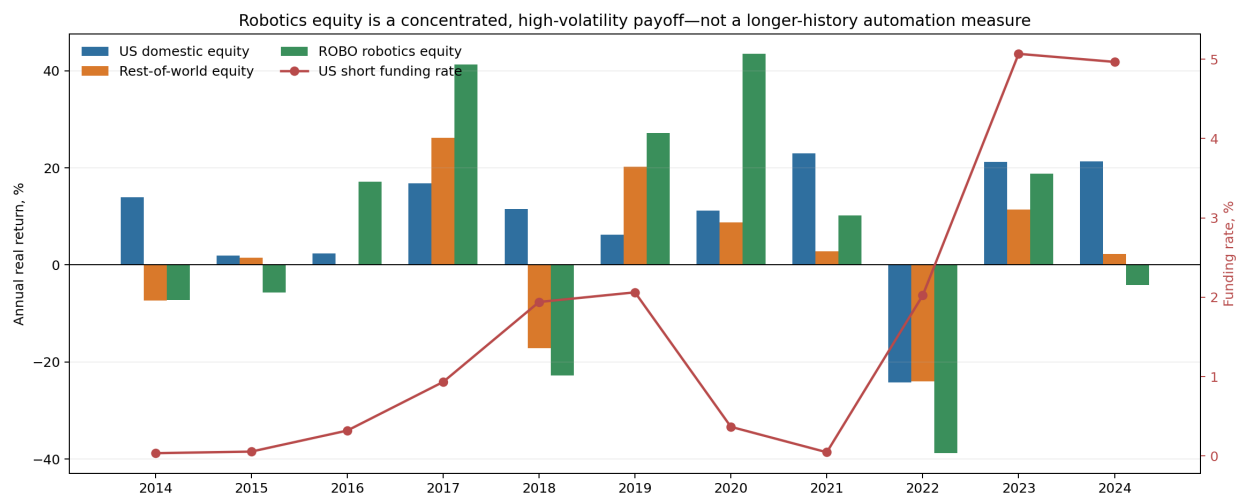


Figure D5: Recent robotics-equity returns and funding conditions. This existing output uses a short common security history. Local-currency returns include exchange-rate and inflation components. It does not estimate an automation premium or an optimal public portfolio.

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